

AI Knowledge Graph - Detailed Tech Stack (Short Table Format)

React Enterprise Dashboard (Milestone 4)

Technology	Version / Model	Purpose
React.js	v18+	Interactive enterprise dashboard UI
Axios	Latest	API communication between frontend & backend
React-Force-Graph-2D	D3-based	Interactive force-directed graph visualization
Recharts	Latest	Pie chart for relationship distribution
FastAPI	Python Async Framework	Backend RAG & graph APIs
Uvicorn	ASGI Server	Run FastAPI server
SentenceTransformer	all-MiniLM-L6-v2 (384-dim)	Semantic embedding generation
Pinecone	Cloud Vector DB	Scalable semantic similarity search
Neo4j	Graph Database (Cypher)	Store nodes & relationships
Ollama	phi3 / mistral	Grounded AI insight generation
Pandas	Latest	Dataset loading & preprocessing
psutil	Latest	CPU & Memory monitoring
dotenv	Latest	Secure environment variable management

Streamlit Graph-Aware RAG (Milestone 3)

Technology	Version / Model	Purpose
Streamlit	Latest	Rapid prototype UI for semantic search
FastAPI	Python Async Framework	RAG backend endpoints
Uvicorn	ASGI Server	Run backend
SentenceTransformer	all-MiniLM-L6-v2 (384-dim)	Query & document embeddings
FAISS	Local ANN Index	In-memory semantic similarity search
Neo4j	Graph Database	Graph filtering & relationship reasoning
Pandas	Latest	Dataset processing
dotenv	Latest	Environment configuration
JSON / AST	Python Libraries	Metadata parsing